

# Experiment 12

## Firewall Configuration & Security Log Analysis

### Lab 12 Objectives:

- Implement firewall rules to block/allow specific traffic and analyze security logs.

**Prerequisites:** The prerequisites of this lab are-

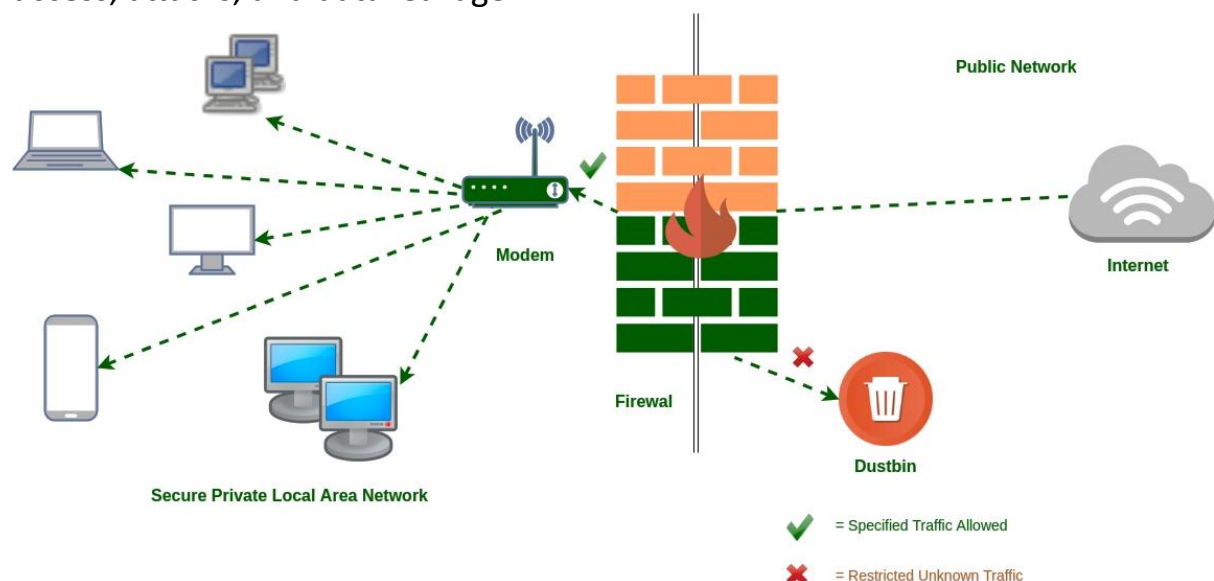
- Basic understanding of firewall concepts and network security.
- Familiarity with IP filtering, ports, and protocol-based rules.

**Outcomes:** By the end of this lab, students will be able to:

- Configure firewall rules to allow or block specific traffic.
- Monitor and analyze firewall security logs.
- Evaluate the effectiveness of firewall policies in securing a network.

### Introduction

A **firewall** is a network security device or software that monitors and controls incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the internet. Firewalls filter traffic to prevent unauthorized access, attacks, and data leakage.



## Experiment Description

### 1. Setting up Firewall:

- A firewall was configured using software tools like **iptables**, **firewalld**, or hardware firewalls like **Cisco ASA**.
- The firewall acts as a gatekeeper by applying filtering rules to network packets.

### 2. Defining Firewall Rules:

- Rules were created to **allow** or **block** traffic based on:
  - **IP addresses** (source or destination)
  - **Port numbers** (e.g., HTTP on port 80, SSH on port 22)
  - **Protocols** (TCP, UDP, ICMP)
- Examples of rules:
  - Allow incoming HTTP traffic on port 80.
  - Block incoming SSH traffic from unknown IPs.
  - Drop ICMP (ping) requests to prevent ping floods.

### 3. Testing Firewall Behavior:

- Attempts were made to connect using allowed and blocked protocols and ports to verify rule enforcement.
- Tools like **nmap** were used for port scanning to test firewall defenses.

### 4. Monitoring Firewall Logs:

- Logs were monitored using tools like **journalctl**, **logwatch**, and **fail2ban**.
- Logs show records of allowed connections, blocked attempts, and possible attack signatures.

### 5. Analyzing Security Logs:

- Log entries were analyzed to detect patterns such as repeated blocked attempts from a single IP (possible brute-force attack).
- Suspicious activity was identified and documented.

### 6. Documentation:

- Firewall rules configuration, commands used, and their outputs were recorded.
- Security log excerpts highlighting blocked and allowed traffic were saved.
- Analysis report was prepared on firewall effectiveness.

## Firewall Concepts and Rule Types

- **Default Policies:** Firewall default action for incoming and outgoing traffic (allow or deny).
- **Allow Rules:** Explicitly permit traffic matching criteria.
- **Deny/Drop Rules:** Block traffic that matches rule conditions.
- **Stateful Inspection:** Tracks connection states to allow returning traffic for established connections.
- **Logging:** Records all allowed or blocked packets based on rules for audit and analysis.

## Tools Used

- **Firewall Software:**
  - **iptables:** Command-line tool for Linux firewall management.
  - **firewalld:** Dynamic firewall daemon with zone support.
  - **UFW (Uncomplicated Firewall):** Simplified iptables interface.
  - **pfSense / Cisco ASA:** Dedicated firewall appliances.
- **Log Analysis Tools:**
  - **journalctl:** For viewing system logs in Linux.
  - **logwatch:** Summarizes log files and detects anomalies.
  - **fail2ban:** Automatically bans IPs with suspicious behavior.
- **Network Tools:**
  - **netstat:** Displays network connections and listening ports.
  - **tcpdump:** Captures network packets.
  - **nmap:** Network port scanner.

## Example Firewall Rules

```
# Default policy to drop all incoming traffic
iptables -P INPUT DROP

# Allow all outgoing traffic
iptables -P OUTPUT ACCEPT

# Allow incoming HTTP traffic on port 80
iptables -A INPUT -p tcp --dport 80 -j ACCEPT

# Allow incoming SSH traffic only from a specific IP
iptables -A INPUT -p tcp -s 192.168.1.100 --dport 22 -j ACCEPT

# Log and drop all other incoming traffic
iptables -A INPUT -j LOG --log-prefix "Dropped Packet: "
iptables -A INPUT -j DROP
```

## Firewall Log Analysis

- Logs typically contain:
  - **Timestamp** of event.
  - **Source IP address** of incoming request.
  - **Destination port and protocol.**
  - **Action taken** (ACCEPT or DROP).
  - Example log excerpt:
- By analyzing logs, we can:
  - Detect repeated unauthorized access attempts.
  - Identify scanning or brute force attacks.
  - Confirm legitimate access is allowed.

## Observations

- Firewall successfully blocked unauthorized access attempts on blocked ports.
- Allowed traffic on open ports passed without interruption.

- Logs clearly showed blocked IP addresses, helping in further security decisions.
- Performance impact due to firewall was negligible for normal traffic levels.

## Security Threat Detection via Firewall Logs

